

Question ID: 9afe2370

Question

The population P of a certain city y years after the last census is modeled by the equation below, where r is a constant and P_0 is the population when $y = 0$.

$$P = P_0(1 + r)^y$$

If during this time the population of the city decreases by a fixed percent each year, which of the following must be true?

Answer

A. $r < -1$

B. $-1 < r < 0$

C. $0 < r < 1$

D. $r > 1$